

AI Workshop Series — Preparation Assignment

AI Expert Team

Estimated time: 4–5 hours

Deadline: before the first technical workshop on September 11

Why you are doing this

During the AI Taskforce workshops, we will use AI coding harnesses to build, test, debug and explain practical data and AI solutions for SMEs. This preparation assignment is your first small end-to-end exercise.

The objective is **not** to deliver a production-ready application or a perfect business recommendation. The objective is to arrive at the first workshop with:

- a working local development environment;
- experience using an AI coding harness to build and debug something tangible;
- a small local application and automated report you can explain; and
- at least one concrete question, failure or uncertainty to discuss together.

You are expected to use your coding harness to install, configure and troubleshoot the stack independently. Different repository structures and implementation choices are expected. Keep track of what you tried when something did not work.

Required stack

Use the following stack for this assignment:

- **AI coding harness:** Pi or your preferred Agentic AI Harness (e.g. Claude Code, Github Copilot)
- **Application:** Python with Flask
- **Place data:** OpenStreetMap, using Nominatim for address lookup and Overpass for venue data
- **Reporting:** Quarto, rendered as a PDF
- **Deployment:** none: the application should run locally

You do **not** need a commercial maps API key for the core assignment. Use OpenStreetMap/Nominatim/Overpass responsibly and at modest volume. Do not repeatedly send identical requests while testing.

Preparation resources

Please watch these resources before or while completing the assignment. They introduce the concepts and tools that we will use in the workshops.

1. What is an AI Harness?

https://www.youtube.com/watch?v=KX_VwNhcFtk&t=30

2. **How an AI Agent works: system prompts, context and skills**
https://www.youtube.com/watch?v=S_oN3vlzpMw
3. **The New Software Development Cycle with Vibe Coding - Google Training PDF** <https://readwise-assets.s3.amazonaws.com/media/wisereads/articles/the-new-sdlc-with-vibe-coding/1317.pdf>
4. **Pi Agent Harness, briefly explained**
<https://www.youtube.com/watch?v=FJxgz5pN4wU>
5. **Quarto basics**
<https://www.youtube.com/watch?v=Q3phTByW138>
6. **Style Quarto Reports with Typst** <https://www.youtube.com/watch?v=idgZjnDIS1s>

Advice / suggestions

Iterate on your result

The first solution produced by an AI coding harness may look convincing or even work for one test case. Do not stop there. Run your application, test more than one address and establishment type, inspect the generated report, and use the results to find bugs and improvements.

Iterate with your coding harness: explain what you observed, provide relevant error messages or output, ask for a focused fix, and verify the change yourself. We are deliberately interested in what did *not* work at first. Keep short notes on the prompts, decisions and issues you encounter so that we can discuss them in the workshop.

How can I get access to LLM APIs?

Your coding harness requires access to at least one LLM provider. If you already have an API key or subscription with a provider such as OpenAI, Anthropic, or Mistral, you are welcome to use it. Please review the provider's pricing, usage policies, and spending controls before getting started.

If you do not already have access, we can provide a limited-use API key for GPT-5.6-luna, a state-of-the-art model with strong coding capabilities. To request a key, contact n.e.jagergallego@hhs.nl or send us a message via the WhatsApp group

To ensure fair access for everyone and keep costs manageable, usage on institution-provided keys may be monitored and subject to quotas or spending limits. Please use these keys responsibly and avoid unnecessary or excessively large requests.

Assignment: HORECA competition lens

Build a small local competition-analysis application for an aspiring HORECA entrepreneur. The entrepreneur wants to investigate a potential location before opening a **café, restaurant** or **hotel**.

The application should use the entered address to find relevant, *mapped* OpenStreetMap establishments nearby. OpenStreetMap data is useful but can be incomplete or out of date; treat it as an indication, not a complete market survey.

Core requirements

Build a local Flask application that allows a user to:

1. Enter an address.

2. Select one establishment category:
 - café;
 - restaurant; or
 - hotel.
3. Search for matching mapped establishments within a **1 km radius** of that address.
4. Display the **five closest** matching establishments. For each result, show:
 - name;
 - type;
 - website, where available; and
 - straight-line distance from the entered address.

If fewer than five matching establishments are found, show all available results and make that clear in the interface.

Automated Quarto report

Add a way to generate a Quarto PDF report from the current analysis. The report should include:

- the entered address, selected category and search radius;
- a clearly formatted table of the five closest establishments;
- the total number of matching mapped establishments found within 1 km;
- the distance to the closest matching establishment;
- an automatically generated, evidence-based indication of local competition; and
- a short limitations statement about incomplete OpenStreetMap metadata and the need to validate results before making a real business decision.

The core competition assessment should be **rule-based and transparent**. It does not need an LLM. Use the number of matching establishments within the radius and the distance to the closest competitor to produce a short conclusion.

You may use the following logic, or define a similarly clear rule yourself:

Observed result	Indicative conclusion
0–1 matching competitor within 1 km	Lower observed direct competition
2–4 matching competitors within 1 km	Moderate observed direct competition
5 or more matching competitors within 1 km	Higher observed direct competition
Closest competitor is under 250 m	Add a note that a direct competitor is very nearby

For example:

Indicative competition assessment: high. The OpenStreetMap query identified 8 mapped restaurants within 1 km of the proposed location. The nearest mapped restaurant is approximately 120 metres away. This suggests that the location is already served by several direct competitors. Validate these results through local field research and a clearer definition of the intended restaurant concept before making a location decision.

Your wording does not need to match this example. It should simply be grounded in the retrieved data and make its limitations clear.

Optional extension: AI-assisted narrative

Once the core assignment works, you may use an LLM to draft a short two- or three-sentence business interpretation for the report.

If you do this:

- include the relevant counts, radius and closest-competitor distance in the prompt;
- label the result as **AI-assisted** in your report or documentation; and
- verify that every factual claim in the text can be traced back to your retrieved data.

Do not use an LLM to replace the rule-based assessment. The aim is to compare a transparent automated rule with an AI-generated narrative and critically assess both.

Tip: to include a simple agent inside your Python application you can install the following skill to your agent harness and steer it to integrate a compatible Pydantic AI agent: <https://pydantic.dev/docs/ai/overview/coding-agent-skills/#cross-agent-agentskillsio>

What to submit

Upload the following to the AI Taskforce Teams channel in a new folder that has your name:

1. Your final Quarto PDF report.
2. A link to your GitHub repository **or** a ZIP file containing your source code.
3. A short reflection in the email or in your README.md that answers:
 - What worked well?
 - What did not work at first?
 - What is one issue you solved by iterating with your coding harness?
 - What is one thing you still do not understand or want to discuss during the workshop?

Responsible use

- Use only public OpenStreetMap data for this assignment.
- Do not upload confidential, personal, client or SME data to an external AI provider.
- Never commit API keys, passwords or other secrets to your repository.
- Treat AI-generated code, summaries and conclusions as drafts: run the application and verify important output yourself.
- Treat the generated report as an exploratory indication, not as a real-world business decision tool.

Definition of done

You are finished when you can demonstrate a locally running application, generate a PDF report for at least one address/category combination, and explain one thing that worked and one thing that required debugging or further investigation.

Do not spend the full 4–5 hours polishing the interface or adding features. A small, working, explainable solution is more valuable than a large but unverified one.